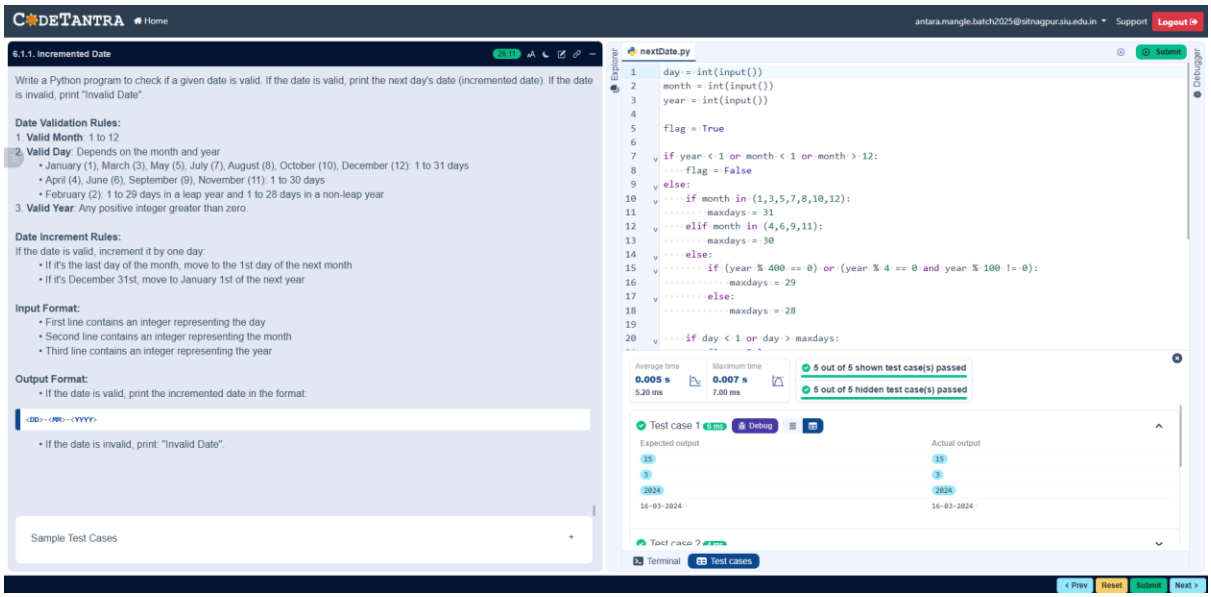
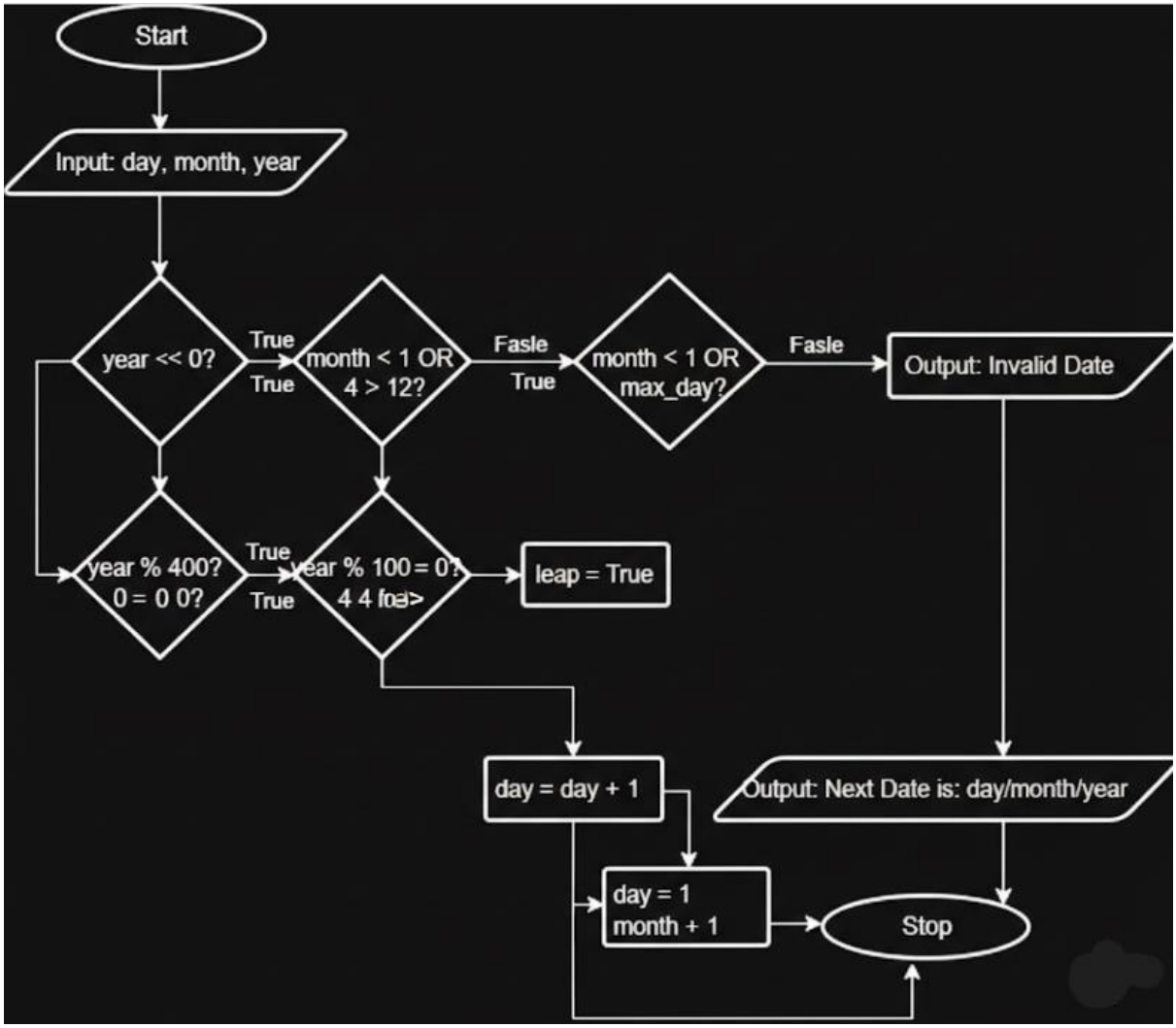
6.1



Algorithm

1. Start.
2. Read day, month, and year from the user.
3. If year ≤ 0 → Print "Invalid Date" → Stop
4. If month < 1 OR month > 12 → Print "Invalid Date" → Stop
5. If (year % 400 == 0) → Set leap = True
6. Else if (year % 100 == 0) → Set leap = False
7. Else if (year % 4 == 0) → Set leap = True
8. Else → Set leap = False
9. If month is 1, 3, 5, 7, 8, 10, or 12 → Set max_day = 31
10. Else if month is 4, 6, 9, or 11 → Set max_day = 30
11. Else if month == 2 → If leap == True → Set max_day = 29
    → Else → Set max_day = 28
12. If day < 1 OR day > max_day → Print "Invalid Date" → Stop
13. Increment day by 1
14. If day > max_day → Set day = 1 → Increment month by 1
15. If month > 12 → Set month = 1 → Increment year by 1
16. Print "Next Date is: day/month/year"
17. Stop.

Flowchart


Start
Input: day, month, year
year << 0?
True
True
month < 1 OR
4 > 12?
Fasle
True
month < 1 OR
max_day?
Fasle
Output: Invalid Date
year % 400?
0 = 0 0?
True
True
year % 100 = 0?
4 4 foe>
leap = True
day = day + 1
Output: Next Date is: day/month/year
day = 1
month + 1
Stop